

Afsana Bhuiyan

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Work Experience

Margin Research

New York, NY

Software Engineer (Contract)

02/2026 – 06/2026

- Built and shipped v1 of Reagent, a SaaS threat detection platform that identifies malicious open-source contributors, delivered to government customers in four months.
- Architected a RESTful API with FastAPI and Pydantic models and engineered an asynchronous task pipeline using Celery and Redis to orchestrate analysis across internal microservices, indexing results in Elasticsearch for search and storing raw artifacts in S3.
- Developed a reusable TypeScript visualization tool for graph-based analysis, supporting multiple data formats and per-node analytics which was adopted across two internal projects.
- Containerized application services with Docker, built a CI/CD pipeline in GitHub Actions, and wrote pytest unit tests with 85% test coverage.

Pieces

New York, NY

Software Engineer Intern

09/2025 – 12/2025

- Built a full-stack Flask and JavaScript internal tool using the Gemini API to generate synthetic patient records for training and testing the company's AI document-summarization product.
- Applied prompt engineering and built a data pipeline to auto-repair malformed LLM responses, enabling reliable generation of 200+ clinical notes per patient.
- Designed patient record data models with physician consultants to ensure clinical realism, and added batch processing with JSON exports for downstream model training.

COOP Careers

New York, NY

Data Analytics Fellow

01/2025 – 06/2025

- Designed Tableau dashboards and analyzed a 508-film dataset using Python, pandas, and NumPy to identify performance trends across markets, franchises, and genres.

Projects

El Principito - Language Learning App | C#, ASP.NET Core (.NET), SQLite, React Native

08/2026 – Present

- Created a reading-based immersive language-learning mobile app, designing RESTful API endpoints documented and tested with OpenAPI (Swagger) and a SQLite database schema to support book retrieval, vocabulary lookup, and learning progress data.
- Developing mobile frontend in React Native and TypeScript (Expo) with tap-to-translate feature.

CareTrack - Patient Admissions Tracker | React, Express.js, PostgreSQL, Docker, Azure

03/2026 – Present

- Built a HIPAA-compliant React, Express, and PostgreSQL platform that tracks patient admissions and automatically creates physician documentation tasks from each admit date.
- Implemented JWT authentication, packaged API as Docker image, and deployed to Azure Container Apps.

Education

Rensselaer Polytechnic Institute | Troy, NY

06/2024

B.S. in Computer Science, Data and AI Specialization

Skills

Languages: Python, TypeScript, JavaScript, SQL, C#, Java, C/C++, HTML/CSS

Frameworks & Libraries: FastAPI, Flask, Pydantic, Node.js, Express.js, ASP.NET Core (.NET), Entity Framework Core, Celery, React, React Native (Expo), pandas, NumPy

Databases & Cloud: PostgreSQL, Redis/Valkey, Elasticsearch, MongoDB, SQLite, AWS (S3), Azure

DevOps & Tools: Docker, Docker Compose, GitHub Actions, Git, Linux, OpenAPI/Swagger, JWT, Pytest, JUnit, Gemini API, Tableau